

Differentiation • rules to remember

Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Instantaneous slope; the limit must exist.

Constant / power

$$\frac{d}{dx}c = 0; \quad \frac{d}{dx}x^n = nx^{n-1}$$

n is constant; use where the real function is differentiable.

Linearity

$$\frac{d}{dx}[af(x) \pm bg(x)] = af'(x) \pm bg'(x)$$

a and b are constants.

Product

$$(uv)' = u'v + uv'$$

Differentiate one factor at a time; then add.

Quotient

$$\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$$

Bottom $\times$ derivative of top, minus top $\times$ derivative of bottom; $v \neq 0$.

Chain

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

Outer derivative $\times$ inner derivative.

Composite power

$$(u^n)' = nu^{n-1}u'$$

Do not forget u' .

Exponential

$$(e^u)' = e^u u'; \quad (a^u)' = a^u \ln(a) u'$$

$a > 0$; $a \neq 1$ for the usual exponential base.

Logarithm

$$(\ln|u|)' = \frac{u'}{u}$$

$u \neq 0$. For $\ln(u)$ itself, $u > 0$.

Other log bases

$$[\log_a(u)]' = \frac{u'}{u \ln(a)}$$

$u > 0$; $a > 0$ and $a \neq 1$.

Inverse function

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

Local differentiable inverse; denominator $\neq 0$. f^{-1} is not $\frac{1}{f}$.

Local approximation

$$\Delta y \approx f'(x)\Delta x$$

Useful for small input changes; check units and curvature.

Trigonometry & integration essentials

Sine / cosine

$$(\sin x)' = \cos x; \quad (\cos x)' = -\sin x$$

Angles must be in radians.

Tangent / cotangent

$$(\tan x)' = \sec^2 x; \quad (\cot x)' = -\csc^2 x$$

Only where each function is defined.

Secant / cosecant

$$(\sec x)' = \sec x \tan x; \quad (\csc x)' = -\csc x \cot x$$

For any inner $u(x)$, also multiply by u' .

Constant / power integral

$$\int c \, dx = cx + C; \quad \int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

Power rule requires $n \neq -1$.

Reciprocal

$$\int \frac{1}{x} \, dx = \ln|x| + C$$

$x \neq 0$; this is the $n = -1$ exception.

Linearity

$$\int (af \pm bg) \, dx = a \int f \, dx \pm b \int g \, dx$$

Use one arbitrary constant in the final answer.

Exponentials

$$\int e^{kx} \, dx = \frac{e^{kx}}{k} + C$$

$k \neq 0$. Also $\int a^x \, dx = \frac{a^x}{\ln(a)} + C$, $a > 0$, $a \neq 1$.

Basic trig integrals

$$\int \sin x \, dx = -\cos x + C; \quad \int \cos x \, dx = \sin x + C$$

Differentiate the answer to check the sign.

More trig integrals

$$\int \sec^2 x \, dx = \tan x + C; \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C; \quad \int \csc x \cot x \, dx = -\csc x + C.$$

Inverse trig patterns

$$\int \frac{dx}{1+x^2} = \arctan x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C \quad \text{for } |x| < 1.$$

Substitution

$$u = g(x), \quad du = g'(x) \, dx$$

$$\int f(g(x))g'(x) \, dx = \int f(u) \, du. \quad \text{Change definite bounds too.}$$

Integration by parts

$$\int u \, dv = uv - \int v \, du$$

Choose u to simplify when differentiated; LIATE is a heuristic.

Definite integrals & choosing a method

Rectangle sum

$$\Delta x = \frac{b-a}{n}; \quad \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \int_a^b f(x) dx$$

For continuous f , the sums converge as the largest interval shrinks to zero.

FTC: accumulation

$$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$$

For continuous f . With upper bound $g(x)$, multiply $f(g(x))$ by $g'(x)$.

FTC: evaluation

$$\int_a^b f(x) dx = A(b) - A(a), \quad A' = f$$

Antiderivative at upper limit minus antiderivative at lower limit.

Recover the original

$$\int f'(x) dx = f(x) + C$$

A starting value fixes C ; differentiation loses that information.

Signed / geometric area

$$\text{Net area} = \int_a^b f(x) dx; \quad \text{Total area} = \int_a^b |f(x)| dx$$

For $a < b$, below-axis contributions are negative in the net integral.

Bounds / split intervals

$$\int_a^b f = -\int_b^a f; \quad \int_a^c f + \int_c^b f = \int_a^b f$$

Same endpoints give zero; split at sign changes for geometric area.

Average value

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$

$b > a$; integral units = vertical units $\times$ horizontal units.

Partial fractions

$$\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$$

Integral = $\ln|x| - \ln|x+1| + C$; avoid $x = 0, -1$.

Trig substitution

$$\sqrt{a^2 - x^2} : \quad x = a \sin \theta$$

$a > 0$; choose a valid angle interval and transform dx . Other root forms need other substitutions.

Numerical integration

$$\text{Area} \approx \sum_i \frac{(x_{i+1} - x_i)(y_i + y_{i+1})}{2}$$

For sampled data or no simple antiderivative; finer spacing helps for smooth curves.

Method selection

Simplify $\rightarrow$ basic rule $\rightarrow$ substitution $\rightarrow$ parts or fractions

Not every integral has an elementary antiderivative. Check your result by differentiating.

Quality & reliability • application card

Process sensitivity

$$\frac{dy}{dx}; \quad \Delta y \approx \frac{dy}{dx} \Delta x$$

Example: slope 0.4 MPa per °C $\times 5^\circ\text{C} \approx 2\text{MPa}$ locally; illustrative, not a fitted process model.

Accumulated defects

$$\text{Expected defects} = \int_a^b r(t) dt$$

$r(t)$ is expected defects per hour. Do not integrate a defect fraction without throughput.

Quality probability

$$P(\text{LSL} \leq X \leq \text{USL}) = \int_{\text{LSL}}^{\text{USL}} p(x) dx$$

Continuous density p ; outside-spec probability = 1 minus this area.

Lifetime distribution

$$F(t) = P(T \leq t); \quad R(t) = 1 - F(t)$$

For continuous nonnegative lifetime T : $F(t) = \int_0^t f(s) ds$.

Density and survival

$$f(t) = F'(t) = -R'(t)$$

Probability in $[a, b] = \int_a^b f(t) dt = R(a) - R(b)$. Density is not point probability.

Hazard rate

$$h(t) = \frac{f(t)}{R(t)} = -\frac{d}{dt} \ln R(t)$$

$R(t) > 0$. Units: inverse time. Conditional failure probability over small $\Delta t \approx h(t)\Delta t$.

Cumulative hazard

$$H(t) = \int_0^t h(s) ds; \quad R(t) = e^{-H(t)}$$

Assumes a continuous lifetime with $R(0) = 1$.

Mean time to failure

$$\text{MTTF} = E[T] = \int_0^\infty R(t) dt$$

Nonnegative lifetime; finite mean if the integral converges. Not repairable-system MTBF.

Exponential model

$$R(t) = e^{-\lambda t}; \quad f(t) = \lambda e^{-\lambda t}; \quad h(t) = \lambda$$

$\lambda > 0$; $t \geq 0$; constant hazard assumption. $F(t) = 1 - e^{-\lambda t}$; $\text{MTTF} = \frac{1}{\lambda}$.

Exponential example

$$\lambda = 0.002 \text{ h}^{-1}; \quad R(100 \text{ h}) = e^{-0.2} \approx 0.8187$$

$F(100\text{h}) \approx 0.1813$; $\text{MTTF} = 500\text{h}$. Hazard is $\frac{0.002}{\text{h}}$, not 18.13%.

Remember

Slope $\rightarrow$ rate; area $\rightarrow$ accumulation

Keep C; use radians; include inner derivatives; check denominators and model assumptions.